Improved Analytical Modeling of Permanent Magnet Leakage Flux in Design of the Coreless Axial Flux Permanent Magnet Generator Amélioration du Model Analytique de Fuite du Flux dans la Procédure de Conception d'un Générateur sans Noyau en Utilisant l'Aimant Permanent à Flux Axial

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***Abstract*— This paper presents an improved magnetic circuit model of a coreless axial flux permanent magnet (AFPM) machine for the evaluation of its air-gap leakage flux. In the coreless AFPM machine, the stator axial length is one of the effective parameters in the calculation of the air-gap leakage flux factor. It is important to correctly update the leakage flux factor value corresponding to the variation of machine dimensions during the optimal design procedure. In this regard, first, an improved magnetic equivalent model of AFPM is achieved by using a nonlinear iterative algorithm to consider the magnetic saturation of rotor back iron cores. Then, the distribution of the flux in all parts of the machine is obtained. In other words, a quasi-3-D model of the coreless AFPM machine is considered based on the sizing equations to analytically obtain the air-gap leakage flux factor in terms of machine dimensions. Finally, the 3-D finite-element model of the machine is prepared to confirm the validity of analytical results. Very little difference between parameter values illustrates the accuracy of the proposed model.**

***Résumé*— Un modèle amélioré du circuit magnétique d'une machine à Aimant Permanent à Flux Axial (APFA) est présenté dans cet article, pour une évaluation de la fuite du flux dans l'entrefer. Dans la machine à APFA sans noyau, la longueur axiale du stator est l'un des paramètres cruciaux dans le calcul de l'entrefer. Il est primordial de mettre à jour la valeur du facteur de fuite du flux correspondant à la variation des dimensions de la machine pendant la procédure de conception pour avoir un résultat optimal. À cet égard, en premier lieu, un modèle amélioré magnétique de l'APFA sans noyau est développé en utilisant un algorithme itératif non linéaire pour tenir compte de la saturation magnétique du rotor. Par la suite, la répartition du flux dans toutes les parties de la machine est obtenue. En d'autres termes, un modèle quasi-3-D de la machine sans noyau APFA est considéré sur la base des équations de dimensionnement pour obtenir analytiquement le facteur de fuite du flux dans l'entrefer, considérant la dimension de la machine électrique. Finalement, le modèle à élément fini en 3-D de la machine est utilisé pour confirmer la validité des résultats analytiques. Le peu de disparité entre les valeurs des paramètres obtenus illustre la précision du modèle proposé.**

***Index Terms*— Axial flux permanent magnet (AFPM) generator, coreless machine, finite-element analysis (FEA), leakage flux modeling, nonlinear algorithm, quasi-3-D.**

NOMENCLATURE

$A_{g,i}$, $A_{\mathrm{pm},i}$	Flux passing area of air gap and magnet in the ith layer.
$A_{r1,i}$, $A_{r2,i}$	Average flux passing areas of the corresponding iron part in the ith layer.
B	Magnetic flux density vector.
$B_{r1,i}$, $B_{r2,i}$	Flux densities of the corresponding iron part in the ith layer.
$D_{\mathrm{av},i}$	Average diameter of the ith layer.
D_{in}, D_{out}	Inner diameter and outer diameter of the machine.
F_{pm}	Equivalent MMF of the PM.
L_g	Effective air-gap length.
L_i	Depth of each layer.
$L_{m,i}$	Maximum radius of flux path in the ith layer.
L_s, L_{pm}, L_r	Stator, PM, and rotor axial length.
N	Total number of layers.
N_c	Number of coil turns.
P_{rated}	Rated power.
$P_{\mathrm{mr},i}$, $P_{\mathrm{mm},i}$	Leakage permeance of magnet-to-rotor and magnet-to-magnet in the ith layer.

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Q_c	Number of stator coils.
$R_{g,i}$, $R_{\mathrm{pm},i}$, $R_{r,i}$	Air gap, PM, and rotor back iron reluctances of the ith layer.
$R_{\mathrm{mr},i}$, $R_{\mathrm{mm},i}$	Magnet-to-rotor and magnet-to-magnet leakage flux reluctances of the ith layer.
$R_{r1,i}$, $R_{r2,i}$	Reluctances of the corresponding iron part in the ith layer.
V_{rated}	Rated phase voltage.
d	Damping factor.
dA_i	Cross-sectional area of each differential permeance.
g	Air-gap length.
i	Number of each layer.
k	Number of iterations.
k_f	Magnet width factor.
k_{pm}	Air-gap leakage flux factor.
n_{rated}	Rated speed.
p	Number of pole pairs.
$w_{\mathrm{pm},i}$	PM width in the ith layer.
x	Radius of flux path.
Δk_{pm}	Error of air-gap leakage flux factor.
$\Delta\Phi_g$	Error of air-gap flux.
$\Delta\Phi_m$	Error of flux leaving the magnet.
$\Phi_{1,i}$, $\Phi_{2,i}$, $\Phi_{3,i}$, $\Phi_{4,i}$	Fluxes of main loops.
Φ_g, Φ_m	Air-gap flux and flux leaving the magnet for a pole pitch.
$\Phi_{g,i}$, $\Phi_{m,i}$, $\Phi_{r,i}$	Air-gap flux, flux leaving the magnet, and rotor back iron flux of ith layer.
$\Phi_{\mathrm{mr},i}$, $\Phi_{\mathrm{mm},i}$	Magnet-to-rotor and magnet-to-magnet leakage flux of ith layer.
$\alpha_{p,i}$	PM width to pole pitch ratio of the ith layer.
μ_0	Air permeability.
$\mu_{f1,i}$, $\mu_{f2,i}$	Relative permeability's of the rotor iron parts in the ith layer.
μ_r	Relative permeability of PM.
$\tau_{p,i}$	Pole pitch in the ith layer.

I. Introduction

RECENTLY, axial flux permanent magnet (AFPM) machines have been used in a wide variety of applications, such as electrical vehicles, wind turbines, and industrial equipment. Since a large number of poles can be accommodated, these machines are ideal for direct-coupled and low-speed applications [1]. Depending on the application and operating environment, the stators of AFPM machines may have ferromagnetic cores [2] or be coreless [3]. Coreless stator AFPM machines exhibit higher efficiencies than other types, due to the absence of core losses; also, there is no attractive force between the stator and the rotor. Other advantages of these machines include lightweight, small size, simple construction, and no torque pulsation, which is important in low-speed applications, such as direct drive small size wind turbines. On the other hand, because of the increased nonmagnetic air-gap length, a coreless machine uses more PM material in comparison with the conventional types, which include ferromagnetic stator core.

In the past two decades, researchers have considered different methods to improve the analytical design procedure of AFPM machines [4]. An analytical design of AFPM machines has been studied in [5] and [6] based on general sizing equations. In [7], the design and the electromagnetic analysis of AFPM machines with ferromagnetic cores are investigated for slotted and nonslotted Torus types. Bumby *et al.* [8] presented an analytical procedure for the calculation of magnetic fields in an AFPM generator by means of current sheet models. They also used this model to design and prototype of a coreless AFPM generator directly coupled to the wind turbine [3]. An optimal design of a coreless stator AFPM generator has been proposed according to a hybrid method in [9]. The method employs a combination of analytical analysis and finite-element analysis (FEA). Chan *et al.* [10] presented a 3-D finite-element model (FEM) of a coreless AFPM generator in comparison with a 2-D FEM. An improved design of a 30-kW AFPM generator using genetic algorithm (GA) for minimum active material cost was presented in [11]. The generator consists of two parallel-connected stators with ferromagnetic core and one inner rotor. Mahmoudi *et al.* [12]–[14] evaluated an optimized AFPM motor by GA for maximum efficiency and power density with reduced torque ripple.

The amount of PM leakage flux is one of the important parameters in the AFPM machine design procedure, especially in the coreless AFPM machines. In a coreless AFPM machine with two rotors and one coreless stator, the effective air-gap length of the machine, which is an effective parameter in the value of PM leakage flux, varies by the stator axial length variation; thus, in this case, the leakage flux factor and the PM axial length are changed. But, in the iron-cored machines, stator coils are placed in the slots, and therefore, the air-gap length and the leakage flux factor value do not change by the stator axial length variation. In general, in an iron-cored machine with a constant air-gap length, the value of leakage flux changes in a small range with respect to other parameters variations, whereas in the coreless AFPM machines and especially in an iterative design procedure like optimal design, by the variation of stator coil axial length and other machine parameters, the value of leakage flux factor changes in a larger range and, thus, requires accurate calculation.

In [15], the leakage flux model has been derived for radial flux interior PM machines. Qu and Lipo [16] have regarded the analysis and modeling of PM leakage flux for the iron-cored AFPM machines without considering the rotor back iron reluctances and iron saturation. For iterative design procedure of linear PM motors, an improved magnetic equivalent circuit model is established in [17]. In [18], an improved analytical model for the electromagnetic analysis of a coreless AFPM machine is presented, but the amount of leakage flux and the calculation method is not specified obviously. The leakage flux consideration in the modeling of a high-speed coreless AFPM generator is presented in [19] and [20]; however, the model has not been validated for various values of design dimensions and the consideration of rotor back iron saturation has not been described well. An analytical design framework for coreless radial flux PM machines is also presented in [21], in which the saturation of rotor back iron has been considered.

In the aforementioned research, PM leakage flux computation was considered in the AFPM machine with stator core; hence, a complete analytical model of PM leakage flux has not been presented for the coreless AFPM machine. In addition, in recent studies, the leakage flux calculation was developed in the mean radius of the machine without enough accuracy. In the axial flux machine, because of the intrinsic 3-D geometry of the machine model, an analytical analysis in the average radius of the machine does not possess adequate accuracy. The 3-D FEM analysis is time-consuming and cannot be used in the early stages of a machine design. An analytical model of the AFPM machine using quasi-3-D computation with two stators and one inner rotor has been studied in [22]. In [23], an improved electromagnetic field analysis of the AFPM machine is performed, and the equivalent circuit parameters are achieved.

In this paper, a comprehensive magnetic equivalent circuit model of coreless AFPM machines is provided for accurate computing of PM leakage flux factor based on the quasi-3-D model of the machine. The numerical results about the value of PM leakage flux are presented, which have not been considered in the previous works completely, in the case of coreless AFPM machines. The magnet-to-rotor and magnet-to-magnet leakage flux calculations are analytically considered in terms of machine dimensions and PM material properties. The magnetic saturation of rotor back iron cores is considered, and a nonlinear iterative procedure is used to update the permeability of saturable parts. Although the consideration of saturation factor in the rotor disks has been considered in the case of radial flux machines, but this idea has not been implemented for the coreless AFPM machines before. In order to evaluate the validity of the analytical results, a 3-D FEM of the machine was investigated and analyzed using Maxwell 3-D software for different values of magnet length, stator length, and magnet width to pole pitch ratio. The comparison between the analytical and FEM results verifies the accuracy of the proposed model.

II. Magnetic Circuit Model of Coreless AFPM Machine

The AFPM machine can be considered as a certain number of layers, each of them found as an individual machine with differential radial lengths. Each layer is used to develop a 2-D model of the machine with an average radius of the selected layer. Fig. 1 shows the 3-D model transformation to a corresponding 2-D model of a coreless AFPM machine with two rotors and one inner coreless stator. Trapezoidal PMs are placed on the rotor surface, and the concentrated winding coils in the stator are held together using composite material of epoxy resin. The average diameter of the ith layer can be expressed as (1) in the quasi-3-D computation model

$$D_{\text{av},i} = D_{\text{out}} - (2i-1)\left(\frac{D_{\text{out}} - D_{\text{in}}}{2N}\right) \tag{1}$$

where D_{out} and D_{in} are the outer and inner diameters of the machine, respectively, and N is the number of layers. The pole pitch ($\tau_{p,i}$) and the PM width ($w_{\text{pm},i}$) for each computation

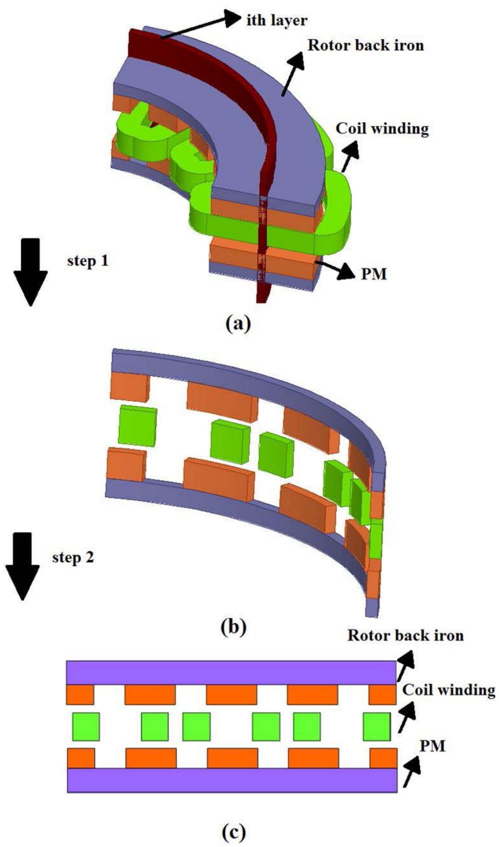


Fig. 1. 3-D to 2-D transformation of the coreless AFPM machine in the quasi-3-D computation. (a) 3-D geometry. (b) Selected layer. (c) 2-D model.

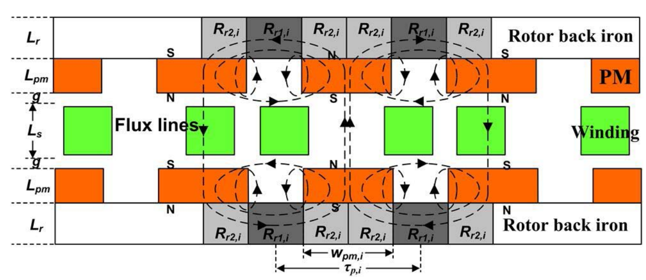


Fig. 2. Simple 2-D structure of coreless AFPM machine in the mean radius of the specified layer.

layer are also expressed as follows:

$$\tau_{p,i} = \frac{\pi D_{\text{av},i}}{2p} \tag{2}$$

$$w_{\text{pm},i} = \alpha_{p,i}\tau_{p,i} \tag{3}$$

where $\alpha_{p,i}$ is the magnet width to pole pitch ratio and p is the number of pole pairs. The magnetic circuit modeling needs to illustrate the poles arrangement and flux paths. For this purpose, Fig. 2 shows the 2-D view of a coreless AFPM generator in the mean radius of the specified layer, in which flux paths are shown with dashed lines. In the coreless AFPM machines, the zigzag leakage flux is eliminated because of the nonslotted stator structure and large effective air gap in comparison with slotted ferromagnetic stator cores [16].

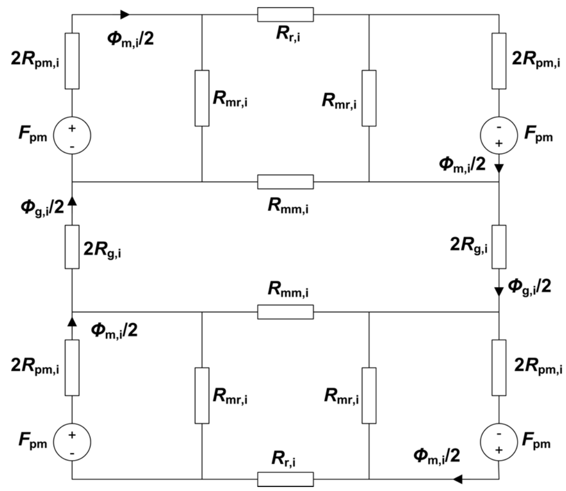


Fig. 3. Magnetic equivalent circuit for the structure in Fig. 2 (one half of a pole pair).

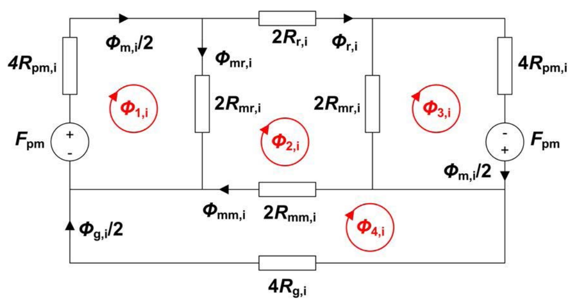


Fig. 4. Simple form of magnetic equivalent circuit in Fig. 3.

In Fig. 2, L_g, L_s, L_{pm}, L_r, and g are, respectively, the effective air-gap length between two PMs, stator axial length, PM axial length, rotor yoke length, and the air-gap length. Based on AFPM modeling in Fig. 1 and flux paths in Fig. 2, the magnetic equivalent circuit of the ith layer for one magnet pole is derived and shown in Fig. 3, where $R_{g,i}$, $R_{r,i}$, and $R_{\text{pm},i}$ are the reluctances of the air gap, rotor back iron, and magnet in layer i, respectively. $R_{\text{mr},i}$ and $R_{\text{mm},i}$ are the corresponding reluctances of the magnet-to-rotor leakage flux, and magnet-to-magnet leakage flux, respectively. Because of the large effective air-gap length, the magnetic field of the stator winding current is neglected.

With the combination of reluctances, the magnetic equivalent circuit can be reduced, as shown in Fig. 4. $\Phi_{m,i}$, $\Phi_{g,i}$, $\Phi_{\text{mr},i}$, $\Phi_{\text{mm},i}$, and $\Phi_{r,i}$ are the flux leaving the magnet, air-gap flux, magnet-to-rotor leakage flux, magnet-to-magnet leakage flux, and rotor back iron flux in layer i. The air gap and magnet reluctances can be obtained from

$$R_{g,i} = \frac{L_g}{\mu_0 A_{g,i}} \tag{4}$$

$$R_{\text{pm},i} = \frac{L_{\text{pm}}}{\mu_0 \mu_r A_{\text{pm},i}} \tag{5}$$

where

$$A_{g,i} = \tau_{p,i}\left(\frac{D_{\text{out}} - D_{\text{in}}}{2N}\right) \tag{6}$$

$$A_{\text{pm},i} = \alpha_{p,i} A_{g,i} \tag{7}$$

$$L_g = L_s + 2g \tag{8}$$

where $A_{g,i}$ and $A_{\text{pm},i}$ are the flux passing area of the air gap and magnet in computation layer i, respectively.

For accurate calculations and according to the variation of rotor yokes flux density behind the PMs and interpolar regions, the rotor back iron reluctances are separated into two parts, as shown in Fig. 2, and defined as follows:

$$R_{r,i} = R_{r1,i} + 2R_{r2,i}. \tag{9}$$

Reluctances behind the interpolar regions ($R_{r1,i}$) and behind the PMs ($R_{r2,i}$) are obtained from

$$R_{r1,i} = \frac{\tau_{p,i} - w_{\text{pm},i}}{\mu_0 \mu_{f1,i} A_{r1,i}} \tag{10}$$

$$R_{r2,i} = \frac{0.5 w_{\text{pm},i}}{\mu_0 \mu_{f2,i}(A_{r1,i} + A_{r2,i})/2} \tag{11}$$

where $\mu_{f1,i}$ and $\mu_{f2,i}$ are the relative permeabilities of the rotor iron parts in the corresponding layer i, which are the nonlinear function of flux density and determined through the B–H curve of the rotor iron. $A_{r1,i}$ and $A_{r2,i}$ are the average flux passing areas of the rotor back iron behind the interpolar and PM regions, respectively, given by

$$A_{r1,i} = L_r\left(\frac{D_{\text{out}} - D_{\text{in}}}{2N}\right) \tag{12}$$

$$A_{r2,i} = 0.5 w_{\text{pm},i}\left(\frac{D_{\text{out}} - D_{\text{in}}}{2N}\right). \tag{13}$$

Finally, after the calculations of the leakage reluctances $R_{\text{mr},i}$ and $R_{\text{mm},i}$, which are fully discussed in Section III, the fluxes of magnetic equivalent circuit can be determined from (14) and (15), shown at the bottom of this page.

$$\begin{bmatrix} \varphi_{1,i} \\ \varphi_{2,i} \\ \varphi_{3,i} \\ \varphi_{4,i} \end{bmatrix} = \begin{bmatrix} 4R_{\text{pm},i} + 2R_{\text{mr},i} & -2R_{\text{mr},i} & 0 & 0 \\ -2R_{\text{mr},i} & 4R_{\text{mr},i} + 2R_{\text{mm},i} + 2R_{r,i} & -2R_{\text{mr},i} & -2R_{\text{mm},i} \\ 0 & -2R_{\text{mr},i} & 4R_{\text{pm},i} + 2R_{\text{mr},i} & 0 \\ 0 & -2R_{\text{mm},i} & 0 & 4R_{g,i} + 2R_{\text{mm},i} \end{bmatrix}^{-1} \begin{bmatrix} F_{\text{pm}} \\ 0 \\ F_{\text{pm}} \\ 0 \end{bmatrix} \tag{14}$$

$$\begin{bmatrix} \varphi_{m,i} \\ \varphi_{g,i} \\ \varphi_{\text{mr},i} \\ \varphi_{\text{mm},i} \\ \varphi_{r,i} \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 \\ 1 & -1 & 0 & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} \varphi_{1,i} \\ \varphi_{2,i} \\ \varphi_{3,i} \\ \varphi_{4,i} \end{bmatrix} \tag{15}$$

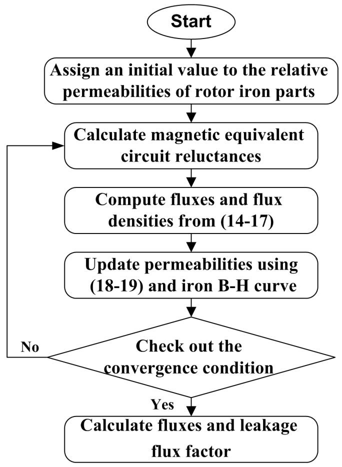


Fig. 5. Nonlinear algorithm of flux and leakage flux calculation.

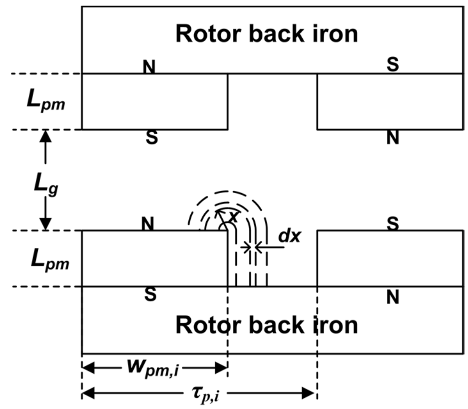


Fig. 6. Permeance model of magnet-to-rotor leakage flux for each layer.

At first, the values of fluxes $\Phi_{1,i}$, $\Phi_{2,i}$, $\Phi_{3,i}$, and $\Phi_{4,i}$ for each main loop are calculated by applying Kirchhoff's Voltage Law to the loops of magnetic circuit model shown in Fig. 4. Then, the values of $\Phi_{m,i}$, $\Phi_{g,i}$, $\Phi_{\mathrm{mr},i}$, $\Phi_{\mathrm{mm},i}$, and $\Phi_{r,i}$ are calculated through (15).

The amount of relative permeability of iron parts, as shown in Fig. 5, is updated in a nonlinear iterative procedure using the B–H curve of the rotor iron. First, the reluctance matrix of the magnetic equivalent circuit model is developed and solved by assigning an initial value to the relative permeability of iron parts. Then, the flux densities in the saturable parts are computed from (16) and (17), and the new permeabilities are updated using B–H curve and (18) and (19) [21]

$$B_{r1,i} = \frac{\varphi_{r,i}}{A_{r1,i}} \tag{16}$$

$$B_{r2,i} = \frac{\varphi_{r,i}}{(A_{r1,i} + A_{r2,i})/2} \tag{17}$$

$$\hat{\mu}_{f,i}^{(k)} = \frac{B_{r,i}^{(k-1)}}{\mu_0 H_{r,i}^{(k-1)}} \tag{18}$$

$$\mu_{f,i}^{(k)} = \left[\hat{\mu}_{f,i}^{(k)}\right]^d \left[\mu_{f,i}^{(k-1)}\right]^{(1-d)} \tag{19}$$

where k is the number of iterations and d is the damping factor, which is set to 0.1 herein. The iteration continues until the convergence condition is satisfied for all permeabilities as follows [21]:

$$\left| \frac{\left(\mu_{f,i}^{(k)} - \mu_{f,i}^{(k-1)}\right)}{\mu_{f,i}^{(k-1)}} \right| \leq 0.01. \tag{20}$$

As can be seen, the flux distribution inside the generator is achieved based on the magnetic equivalent circuit model of the machine and the reluctance of flux paths.

III. Leakage Flux Calculation

As mentioned in Section II, the calculations of the leakage reluctances $R_{\mathrm{mr},i}$ and $R_{\mathrm{mm},i}$, and the leakage flux factor, which is a much effective parameter in the design procedure, is taken into account in this section.

A. Magnet-to-Rotor Leakage Flux

The magnet-to-rotor reluctance for each computation layer ($R_{\mathrm{mr},i}$) can be achieved by computing the leakage permeance between the magnet and the rotor back iron. In [16] and [24], the circular arc straightline model has been used for computing leakage permeance in the air gap of an AFPM machine with a stator core. In the case of coreless stators, according to the large effective air-gap length between opposite magnets on the two rotors, the obtained equations are somewhat different. In Fig. 6, a circular arc straight-line model of magnet-to-rotor leakage flux is shown (dashed lines). The leakage permeance between magnet and rotor for layer i ($P_{\mathrm{mr},i}$) is an infinite sum of differential width permeances, each of length $L_{\mathrm{pm}} + \pi x$, which is expressed as

$$P_{\mathrm{mr},i} = \sum \frac{\mu_0 dA_i}{l} = \sum \frac{\mu_0 L_i dx}{L_{\mathrm{pm}} + \pi x} \tag{21}$$

where $dA_i = L_i dx$ is the cross-sectional area of each differential permeance and L_i is the depth of each layer. Because this equation involves the sum of differential elements, its solution is given by the following integral for each layer:

$$P_{\mathrm{mr},i} = \int_0^{L_{m,i}} \frac{\mu_0 L_i}{L_{\mathrm{pm}} + \pi x} dx = \frac{\mu_0 L_i}{\pi} \ln\left(1 + \frac{\pi L_{m,i}}{L_{\mathrm{pm}}}\right). \tag{22}$$

In (22), $L_{m,i}$ can be fixed as a geometric constraint. It is chosen to be the minimum value of half the effective air-gap length and half the gap between two adjacent PMs, as shown in (23). Due to the reverse relation between reluctance and permeance, the magnet-to-rotor leakage reluctance can be easily calculated

$$L_{m,i} = \min\left(\frac{L_g}{2}, \frac{\tau_{p,i} - w_{\mathrm{pm},i}}{2}\right). \tag{23}$$

B. Magnet-to-Magnet Leakage Flux

Similarly, the magnet-to-magnet reluctance for each computation layer ($R_{\mathrm{mm},i}$) can be achieved by computing the leakage permeance between two magnets. As shown in Fig. 7, the circular arc straightline model is used to compute the permeance between two adjacent magnets. Leakage permeance between two adjacent magnets for layer $i (P_{\mathrm{mm},i})$ is an

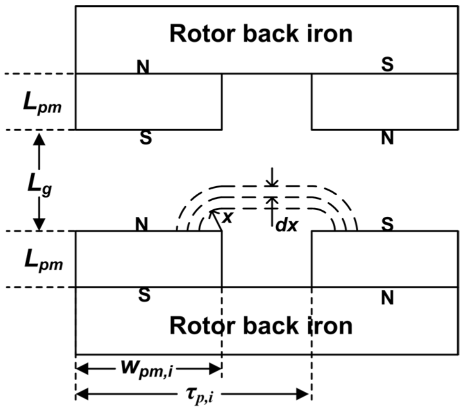


Fig. 7. Permeance model of magnet-to-magnet leakage flux for each layer.

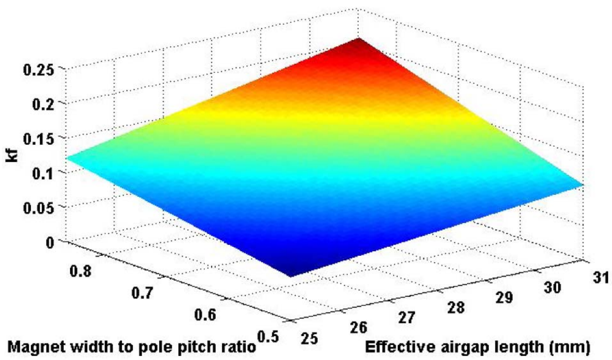


Fig. 8. Surface fitting function of k_f.

infinite sum of differential width permeances, each of length $(\tau_{p,i} - w_{\mathrm{pm},i}) + \pi x$, and can be obtained from

$$P_{\mathrm{mm},i} = \int_0^{k_f w_{\mathrm{pm},i}} \frac{\mu_0 L_i}{(\tau_{p,i} - w_{\mathrm{pm},i}) + \pi x} dx = \frac{\mu_0 L_i}{\pi} \ln\left(1 + \frac{\pi k_f w_{\mathrm{pm},i}}{(\tau_{p,i} - w_{\mathrm{pm},i})}\right). \quad (24)$$

In (24), the upper limit of the integral can be fixed as a geometric constraint. In the case of AFPM machines with a stator core [16], because of the small air-gap length, the upper limit of the integral is chosen equal to the air-gap length. In [21], the value of this parameter is set to $L_g/2$ for the coreless radial flux PM machine. In this paper, because of the large effective air-gap length, the upper limit of the integral is chosen as a fraction of the magnet width ($k_f w_{\mathrm{pm},i}$). In order to precise calculation of the magnet-to-magnet leakage flux, k_f is considered as a surface fitting function of two parameters, effective air-gap length (L_g) and magnet width to pole pitch ratio ($\alpha_{p,i}$), which is calculated by the 2-D FEA of the generator and shown in Fig. 8.

In the same way, the magnet-to-magnet reluctance can be calculated from its permeance ($R_{\mathrm{mm},i} = 1/P_{\mathrm{mm},i}$).

C. Air-Gap Leakage Flux Factor

In the design of coreless AFPM machines, proper consideration of the air-gap leakage flux factor is very important. The use of the coreless stator increases the effective length of the air gap that also causes the PM leakage flux to be increased. At the design stage, with the change of stator winding dimensions, the effective length of the air gap changed, and therefore, the amount of leakage flux changed. It is very important to use an iterative design algorithm, in which the leakage flux factor is updated like other machine dimensions and parameters.

The air-gap leakage flux factor can be obtained as the ratio of the air-gap flux to the flux leaving the magnet. Equation (25) analytically shows the air-gap leakage flux factor in terms of machine dimensions and magnetic material properties. $\Phi_{m,i}$ and $\Phi_{g,i}$ are computed from (15) and updated through the nonlinear iterative algorithm, which is shown in Fig. 5. Total flux is the sum of all layer fluxes in the quasi-3-D model

$$k_{\mathrm{pm}} = \frac{\varphi_g}{\varphi_m} = \frac{\sum_{i=1}^{N} \varphi_{g,i}}{\sum_{i=1}^{N} \varphi_{m,i}}. \quad (25)$$

In Sections II and III an analytical model for the PM leakage flux calculation in the coreless AFPM generator was derived in terms of machine dimensions and material properties. The use of this model in the early stages of a machine design provides an accurate calculation of PM dimensions, which is an important parameter in the design of the coreless AFPM machines. Furthermore, this analytical model can be applied to an iterative design procedure, such as the optimal design of the coreless AFPM generator.

TABLE I
PARAMETERS OF THE CORELESS AFPM MACHINE

Parameter	Symbol	Value
Rated power (kW)	P_{rated}	2
Rated speed (rpm)	n_{rated}	300
Rated phase voltage (V)	V_{rated}	220
Number of pole pairs	p	8
Number of stator coils	Q_{c}	12
Outer diameter of machine (mm)	D_{out}	385
Inner diameter of machine (mm)	D_{in}	222
Physical air gap length (mm)	g	1.5
Number of coil turns	N_{c}	236

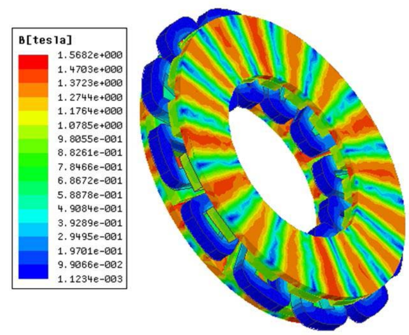


Fig. 9. Magnetic flux density distribution in AFPM generator.

IV. FEM RESULTS AND MODEL VALIDATION

A 3-D FEM of the coreless AFPM generator was analyzed with Maxwell 3-D software to evaluate the validity of the analytical model. Nominal design parameters of the machine are shown in Table I. The 3-D FEM of the generator is shown in Fig. 9, and the magnetic flux density distribution

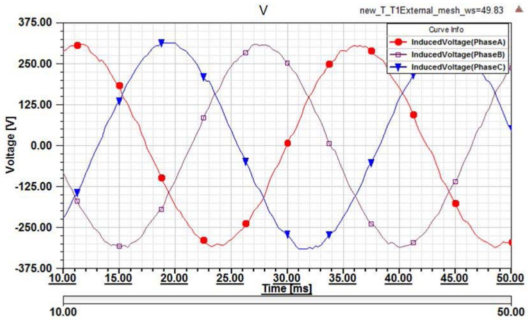


Fig. 10. Three-phase voltages of the AFPM generator under rated load and power factor = 0.9.

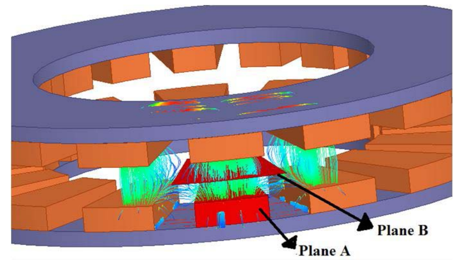


Fig. 11. 3-D flux paths. Plane A: flux leaving the magnet. Plane B: air-gap flux for a pole pitch.

TABLE II
ANALYTICAL RESULTS OF THE PROPOSED MODEL FOR DIFFERENT DIMENSIONS OF THE MACHINE

Case No.	Dimensions			Analytical results		
	L_{pm} (mm)	L_g (mm)	α_i	Φ_m (mWb)	Φ_g (mWb)	k_{pm}
1	10	25.2	0.637	1.625	1.279	0.787
2	11.5	25.2	0.637	1.715	1.354	0.79
3	14.5	25.2	0.637	1.858	1.472	0.793
4	17.5	25.2	0.637	1.967	1.56	0.793
5	16	30.6	0.637	1.861	1.337	0.719
6	16	29	0.637	1.874	1.387	0.74
7	16	27.9	0.637	1.883	1.42	0.754
8	16	26.5	0.637	1.9	1.472	0.775
9	14.5	27.9	0.85	2.289	1.541	0.672
10	14.5	27.9	0.75	2.060	1.490	0.722
11	14.5	27.9	0.637	1.824	1.380	0.756
12	14.5	27.9	0.5	1.528	1.199	0.781

TABLE III
VALUES OF Φ_m, Φ_g, AND k_{pm} OBTAINED BY FEM FOR DIFFERENT DIMENSIONS OF THE MACHINE INTRODUCED IN TABLE II

Case No.	Dimensions			FEM results		
	L_{pm} (mm)	L_g (mm)	α_i	Φ_m (mWb)	Φ_g (mWb)	k_{pm}
1	10	25.2	0.637	1.628	1.274	0.783
2	11.5	25.2	0.637	1.705	1.357	0.796
3	14.5	25.2	0.637	1.845	1.473	0.798
4	17.5	25.2	0.637	1.945	1.568	0.806
5	16	30.6	0.637	1.82	1.278	0.702
6	16	29	0.637	1.808	1.329	0.735
7	16	27.9	0.637	1.832	1.41	0.769
8	16	26.5	0.637	1.879	1.455	0.774
9	14.5	27.9	0.85	2.392	1.621	0.680
10	14.5	27.9	0.75	2.115	1.525	0.721
11	14.5	27.9	0.637	1.757	1.35	0.768
12	14.5	27.9	0.5	1.471	1.152	0.783

is shown in all parts of the generator. The maximum flux density in the rotor yokes occurred behind the interpolar regions corresponding to $R_{r1,i}$, and the portions behind the PMs corresponding to $R_{r2,i}$ operates at lower flux densities. In Fig. 10, the three-phase voltages of generator are displayed under rated load and power factor = 0.9.

In Table II, the analytical results of PM leakage flux are shown based on the proposed model, which is explained in Sections II and III The air-gap flux, flux leaving the magnet, and leakage flux factor are computed for different values of the magnet length, stator length, and magnet width to pole pitch ratio. In each case of Table II, the parameter value of black shaded cells changes and the other two parameters are fixed.

As the results in Table II, show changes in α_i and L_g have the greatest effect on the leakage flux factor value. The reduction of effective air-gap length increases the flux leakage factor (k_{pm}) and, in other words, reduces the leakage flux. Increasing the magnet width to pole pitch ratio (α_i) decreases the distance between adjacent magnets and increases the magnet-to-magnet leakage flux. For α_i changes between 0.85 and 0.5, the value of k_{pm} increased from 0.67 to 0.78.

To confirm the validity of analytical results, FEA has been conducted for each case of Table II. The two surfaces, as shown in Fig. 11, are considered to compute Φ_m and Φ_g in the no-load case. Plane A in Fig. 11, which is considered around all faces of the magnet (except the rotor face), is used to calculate the flux leaving the magnet (Φ_m). The dimensions of this plane are equal to the dimensions of the magnet with little difference. Also, Plane B in Fig. 11, which is located in the middle of the effective air gap, is used to calculate the air-gap flux for a pole pitch (Φ_g). The radial dimension of this plane is equal to $(D_{\text{out}} - D_{\text{in}})/2 + 2w_c$, and the arc size is equal to pole pitch. w_c is the coil width, which is considered in inner and outer diameters of the machine.

The following equation is used to calculate the flux in both cases:

$$\varphi = \iint \vec{B}.\vec{n} dA \tag{26}$$

where B is the magnetic flux density vector and n is the normal vector perpendicular to the plane, in which we want to calculate the magnetic flux through its surface. The 3-D FEM is modified according to the dimensions in Table II, and the results are evaluated for each case. The FEM results are shown in Table III for each case presented in Table II.

The consideration of the rotor back iron reluctances in the magnetic equivalent circuit model of the coreless AFPM machines is one of the important issues that are cited in this paper. To determine the importance of this issue, the analytical results of PM leakage flux for the proposed model have been obtained and compared with the FEM results as well as a previous model, in which the reluctances of rotor back iron and saturation are ignored. The method used in the previous model is similar to the method that has been presented in [16], which is provided again for the coreless machines.

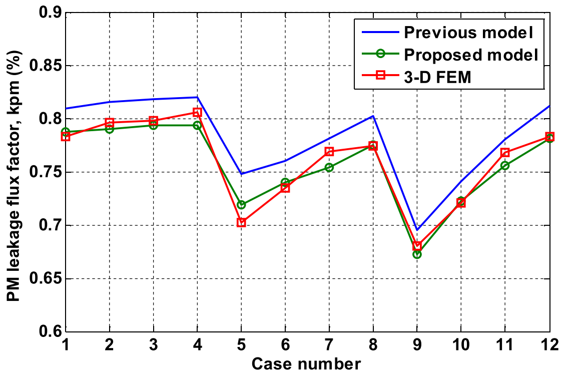


Fig. 12. Analytical and FEM results of PM leakage flux factor for different dimensions of the machine.

TABLE IV
ERROR OF THE PROPOSED MODEL RESULTS IN COMPARISON WITH THOSE FEM RESULTS

Case No.	$\Delta\Phi_{\mathrm{m}}$ (%)	$\Delta\Phi_{\mathrm{g}}$ (%)	Δk_{pm} (%)
1	0.184	-0.392	-0.511
2	-0.587	0.221	0.754
3	-0.705	0.068	0.627
4	-1.131	0.510	1.613
5	-2.253	-4.617	-2.422
6	-3.650	-4.364	-0.680
7	-2.784	-0.709	1.951
8	-1.118	-1.168	-0.129
9	4.306	4.935	1.176
10	2.600	2.295	-0.139
11	-3.813	-2.222	1.563
12	-3.875	-4.080	0.255

The comparison between the results of both analytical models and 3-D FEM is shown in Fig. 12 for different dimensions of the machine introduced in Table II. As it is clear from Fig. 12, the analytical results of the proposed model provide a better agreement with 3-D FEM results in comparison with the results of the previous model.

For more discussion, comparisons between analytical predictions from the proposed model and FEM results for different cases of Tables II and III are shown in Table IV. The errors of analytical results are calculated as follows:

$$\mathrm{Error}(\%) = \frac{(\mathrm{FEM}_{\mathrm{results}} - \mathrm{Analytical}_{\mathrm{results}})}{\mathrm{FEM}_{\mathrm{results}}} \times 100 \equiv \Delta_{\mathrm{results}}. \tag{27}$$

As the results indicate, the results of the proposed quasi-3-D model are in good compatibility with the 3-D FEM results. The errors of Φ_m and Φ_g are <5%, and the leakage flux factor error is <2.5% in all cases. In some cases, even so, the errors of Φ_m and Φ_g are ~4%, but the error of k_{pm} is <1%. The results show that the quasi-3-D analytical model is highly accurate, and therefore, it is an appropriate choice for the optimal design of the machine.

The proposed analytical model of the leakage flux calculation is used to design a coreless AFPM generator with the specifications given in [3]. A 2-kW 500-r/min generator was designed with the open-circuit electromotive force output of 11.16 (V/coil/100 r/min), which is in good agreement with the test result of 11.03 (V/coil/100 r/min) for the prototype generator in [3]. Because of the practical limitations, it is not possible to test more specifically the leakage flux calculation by the experimental data [16].

V. CONCLUSION

In this paper, a quasi-3-D analytical model was developed according to the magnetic equivalent circuit model of the machine for the calculation of PM leakage flux in the coreless AFPM machines. The magnetic saturation of rotor back iron cores was considered, and a nonlinear iterative procedure was used to update the permeability of saturable parts according to the B–H curve of the utilized iron. The analytical results of PM leakage flux have been obtained with and without considering the reluctances of rotor back iron, and have been verified by the 3-D FEA. The results from the proposed model provide a better agreement with the 3-D FEM results in comparison with the results from the previous model.

In the coreless AFPM machines, because of the large effective air-gap length, the air-gap leakage flux has an impressive effect in the design procedure of these machines.

The effect of the three main parameters (magnet length, stator length, and magnet width to pole pitch ratio) is considered in the calculation of leakage flux factor. As the results indicate, changes in α_i and L_g have the greatest effect on the leakage flux factor value. For α_i changes from 0.85 to 0.5, k_{pm} is increased from 0.67 to 0.78. The error of the analytical results was calculated in all considered cases, which is <2.5% for leakage flux factor. A comparison between the analytical predictions and the FEM results shows that the proposed model is highly accurate, and will be appropriate for the purpose of more suitable design of PM machines.

REFERENCES

[1] T. F. Chan and L. L. Lai, "An axial-flux permanent-magnet synchronous generator for a direct-coupled wind-turbine system," *IEEE Trans. Energy Convers.*, vol. 22, no. 1, pp. 86–94, Mar. 2007.

[2] C. C. Chan, "Axial-field electrical machines—Design and applications," *IEEE Trans. Energy Convers.*, vol. EC-2, no. 2, pp. 294–300, Jun. 1987.

[3] J. R. Bumby and R. Martin, "Axial-flux permanent-magnet air-cored generator for small-scale wind turbines," *IEE Proc.-Electr. Power Appl.*, vol. 152, no. 5, pp. 1065–1075, Sep. 2005.

[4] S. Kahourzade, A. Mahmoudi, H. W. Ping, and M. N. Uddin, "A comprehensive review of axial-flux permanent-magnet machines," *Can. J. Electr. Comput. Eng.*, vol. 37, no. 1, pp. 19–33, 2014.

[5] W. Wu, E. Spooner, and B. J. Chalmers, "Design of slotless TORUS generators with reduced voltage regulation," *IEE Proc.-Elect. Power Appl.*, vol. 142, no. 5, pp. 337–343, Sep. 1995.

[6] S. Huang, J. Luo, F. Leonardi, and T. A. Lipo, "A comparison of power density for axial flux machines based on general purpose sizing equations," *IEEE Trans. Energy Convers.*, vol. 14, no. 2, pp. 185–192, Jun. 1999.

[7] M. Aydin, H. Surong, and T. A. Lipo, "Design and 3D electromagnetic field analysis of non-slotted and slotted TORUS type axial flux surface mounted permanent magnet disc machines," in *Proc. IEEE IEMDC*, Jun. 2001, pp. 645–651.

[8] J. R. Bumby, R. Martin, M. A. Mueller, E. Spooner, N. L. Brown, and B. J. Chalmers, "Electromagnetic design of axial-flux permanent magnet machines," *IEE Proc.-Electr. Power Appl.*, vol. 151, no. 2, pp. 151–160, Mar. 2004.

[9] R. J. Wang, M. J. Kamper, K. Van der Westhuizen, and J. F. Gieras, "Optimal design of a coreless stator axial flux permanent-magnet generator," *IEEE Trans. Magn.*, vol. 41, no. 1, pp. 55–64, Jan. 2005.

[10] T. F. Chan, W. Weimin, and L. L. Lai, "Performance of an axial-flux permanent magnet synchronous generator from 3-D finite-element analysis," *IEEE Trans. Energy Convers.*, vol. 25, no. 3, pp. 669–676, Sep. 2010.

[11] N. Rostami, M. R. Feyzi, J. Pyrhonen, A. Parviainen, and V. Behjat, "Genetic algorithm approach for improved design of a variable speed axial-flux permanent-magnet synchronous generator," *IEEE Trans. Magn.*, vol. 48, no. 12, pp. 4860–4865, Dec. 2012.

[12] A. Mahmoudi, S. Kahourzade, N. A. Rahim, W. Hew, and M. N. Uddin, "Design, analysis, and prototyping of a novel-structured solid-rotor-ringed line-start axial-flux permanent-magnet motor," *IEEE Trans. Ind. Electron.*, vol. 61, no. 4, pp. 1722–1723, Apr. 2013.

[13] A. Mahmoudi, S. Kahourzade, N. A. Rahim, and W. P. Hew, "Design, analysis, and prototyping of an axial-flux permanent magnet motor based on genetic algorithm and finite-element analysis," *IEEE Trans. Magn.*, vol. 49, no. 4, pp. 1479–1492, Apr. 2013.

[14] A. Mahmoudi, S. Kahourzade, N. A. Rahim, H. W. Ping, and M. N. Uddin, "Design and prototyping of an optimised axial-flux permanent-magnet synchronous machine," *IET Electr. Power Appl.*, vol. 7, no. 5, pp. 338–349, May 2013.

[15] C.-C. Hwang and Y. H. Cho, "Effects of leakage flux on magnetic fields of interior permanent magnet synchronous motors," *IEEE Trans. Magn.*, vol. 37, no. 4, pp. 3021–3024, Jul. 2001.

[16] R. Qu and T. A. Lipo, "Analysis and modeling of air-gap and zigzag leakage fluxes in a surface-mounted permanent-magnet machine," *IEEE Trans. Ind. Appl.*, vol. 40, no. 1, pp. 121–127, Jan./Feb. 2004.

[17] B. Sheikh-Ghalavand, S. Vaez-Zadeh, and A. H. Isfahani, "An improved magnetic equivalent circuit model for iron-core linear permanent-magnet synchronous motors," *IEEE Trans. Magn.*, vol. 46, no. 1, pp. 112–120, Jan. 2010.

[18] J. Y. Choi, S. H. Lee, K. J. Ko, and S. M. Jang, "Improved analytical model for electromagnetic analysis of axial flux machines with double-sided permanent magnet rotor and coreless stator windings," *IEEE Trans. Magn.*, vol. 47, no. 10, pp. 2760–2763, Oct. 2011.

[19] M. Sadeghierad, H. Lesani, H. Monsef, and A. Darabi, "Leakage flux consideration in modeling of high speed axial flux PM generator," in *Proc. IEEE ICIT*, Apr. 2008, pp. 1–6.

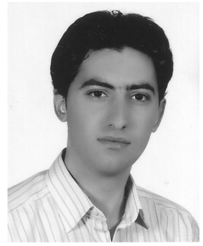
[20] M. Sadeghierad, H. Lesani, H. Monsef, and A. Darabi, "High-speed axial-flux permanent-magnet generator with coreless stator," *Can. J. Electr. Comput. Eng.*, vol. 34, nos. 1–2, pp. 63–67, 2009.

[21] S. Mohammadi and M. Mirsalim, "Analytical design framework for torque and back-EMF optimization, and inductance calculation in double-rotor radial-flux air-cored permanent-magnet synchronous machines," *IEEE Trans. Magn.*, vol. 50, no. 1, Jan. 2014, Art. no. 8200316.

[22] A. Parviainen, M. Niemela, and J. Pyrhonen, "Modeling of axial flux permanent-magnet machines," *IEEE Trans. Ind. Appl.*, vol. 40, no. 5, pp. 1333–1340, Sep. 2004.

[23] S. Y. Sung, J. H. Jeong, Y. S. Park, J. Y. Choi, and S. M. Jang, "Improved analytical modeling of axial flux machine with a double-sided permanent magnet rotor and slotless stator based on an analytical method," *IEEE Trans. Magn.*, vol. 48, no. 11, pp. 2945–2948, Nov. 2012.

[24] D. C. Hanselman, *Brushless Permanent Magnet Motor Design*. New York, NY, USA: McGraw-Hill, 1994.



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